

Exploratory Data Analysis (EDA): A Mathematical and Practical Guide

Contents

1	Introduction to Exploratory Data Analysis	5
2	The Structure of a Dataset	5
3	Types of Variables	6
3.1	Numerical Variables	6
3.2	Continuous Variables	6
3.3	Discrete Variables	6
3.4	Categorical Variables	7
3.5	Ordinal Variables	7
3.6	Binary Variables	7
4	Step 1: Dataset Audit	7
5	Shape of the Dataset	8
6	Data Types	8
7	Unique Values	9
8	Identifiers	9
9	Missing Data	10
10	Mechanisms of Missingness	10
10.1	Missing Completely at Random	10
10.2	Missing at Random	10
10.3	Missing Not at Random	11
11	Missing-Value Visualization	11
12	Handling Missing Numerical Data	11
13	Duplicate Observations	12
14	Descriptive Statistics	12
15	Mean	12
16	Median	13
17	Variance and Standard Deviation	13

18 Quantiles	13
19 Five-Number Summary	13
20 Skewness	14
21 Kurtosis	14
22 Distribution Analysis	14
23 Histogram	15
24 Kernel Density Estimation	15
25 Empirical Cumulative Distribution Function	15
26 Box Plot	16
27 Categorical EDA	16
28 Class Imbalance	17
29 Target Variable Analysis	17
30 Bivariate Analysis	18
31 Covariance	18
32 Pearson Correlation	18
33 Spearman Correlation	19
34 Scatter Plots	19
35 Nonlinear Relationships	20
36 Mutual Information	20
37 Correlation Matrix	20
38 Multicollinearity	21
39 Categorical–Numerical Relationships	21
40 Categorical–Categorical Relationships	22
41 Outlier Analysis	22
42 Z-Score Method	22
43 Robust Outlier Detection	22
44 Multivariate Outliers	23
45 Time-Series EDA	23
46 Trend	23

47 Autocorrelation	24
48 Stationarity	24
49 Differencing	24
50 Feature Scaling	25
51 Min-Max Scaling	25
52 Robust Scaling	25
53 When Scaling Matters	25
54 Transformation of Variables	26
55 Log Transformation	26
56 Power Transformations	26
57 Feature Engineering	26
58 Polynomial Features	27
59 Interaction Effects	27
60 Dimensionality Reduction	27
61 Principal Component Analysis	28
62 t-SNE	28
63 UMAP	28
64 Class Separation	29
65 Target Leakage	29
66 Train-Test Leakage	29
67 Train–Validation–Test Structure	30
68 EDA and Cross-Validation	30
69 Distribution Shift	30
70 Population and Sampling	31
71 Visualization Strategy	31
71.1 Univariate Numerical	31
71.2 Univariate Categorical	31
71.3 Numerical–Numerical	31
71.4 Categorical–Numerical	32
71.5 Categorical–Categorical	32
72 Pair Plot	32

73 Heteroscedasticity	32
74 Residual EDA	33
75 Normality	33
76 Q–Q Plot	33
77 EDA for Regression	34
78 EDA for Classification	34
79 EDA for Time Series	35
80 EDA for Scientific Data	35
81 Dimensional Analysis	36
82 Physical Bounds	36
83 Uncertainty Analysis	36
84 EDA for Neural Operators	37
85 Field Statistics	38
86 Spectral EDA	38
87 Spatial Correlation	38
88 PDE Residual EDA	39
89 Complete EDA Workflow	39
90 A Practical Python EDA Template	40
91 EDA Checklist	43
92 Common EDA Mistakes	45
92.1 Mistake 1: Removing Every Outlier	45
92.2 Mistake 2: Assuming Correlation Means Causation	45
92.3 Mistake 3: Using Only a Correlation Matrix	45
92.4 Mistake 4: Ignoring the Target	45
92.5 Mistake 5: Scaling Before Splitting	45
92.6 Mistake 6: Random Splitting Time Series	45
92.7 Mistake 7: Blindly Removing Missing Rows	45
92.8 Mistake 8: Treating Plots as Proof	46
93 EDA and Model Selection	46
94 Final Summary	46

1 Introduction to Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the systematic process of investigating a dataset before statistical modelling or machine learning.

The purpose of EDA is not merely to generate plots. It is to determine

1. what the dataset contains,
2. how the variables are distributed,
3. whether observations are reliable,
4. whether variables are related,
5. whether missing values exist,
6. whether outliers exist,
7. whether the dataset contains duplicated observations,
8. whether the variables have appropriate scales,
9. whether transformations are necessary,
10. whether assumptions of subsequent models are reasonable,
11. whether data leakage is present,
12. and which modelling approaches are appropriate.

A useful conceptual representation is

$$\text{Raw Data} \rightarrow \text{Data Audit} \rightarrow \text{Univariate Analysis} \rightarrow \text{Bivariate Analysis} \rightarrow \text{Multivariate Analysis} \rightarrow \text{Data Quality}$$

(1)

EDA therefore acts as the bridge between raw observations and statistical or machine-learning modelling.

2 The Structure of a Dataset

Suppose a dataset contains N observations and d features.

The feature matrix is

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{Nd} \end{bmatrix} \in \mathbb{R}^{N \times d}. \quad (2)$$

For supervised learning, there is additionally a target

$$y = [y_1 \quad y_2 \quad \cdots \quad y_N]^T. \quad (3)$$

Thus

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N. \quad (4)$$

The first task of EDA is to determine the mathematical and statistical nature of each x_j and y .

3 Types of Variables

Variables can be classified into several categories.

3.1 Numerical Variables

Numerical variables take quantitative values.

Examples include

$$\text{temperature} = 301.5 \text{ } K, \tag{5}$$

$$\text{pressure} = 1.02 \text{ } atm. \tag{6}$$

They may be continuous or discrete.

3.2 Continuous Variables

A continuous variable can theoretically take values from an interval:

$$X \in [a, b]. \tag{7}$$

Examples include

- temperature,
- mass,
- velocity,
- distance,
- concentration.

3.3 Discrete Variables

Discrete variables take countable values:

$$X \in \{0, 1, 2, \dots\}. \tag{8}$$

Examples include

- number of particles,
- number of defects,
- number of purchases,
- number of events.

3.4 Categorical Variables

A categorical variable takes values from a finite set:

$$X \in \{c_1, c_2, \dots, c_K\}. \quad (9)$$

Examples include

- material type,
- species,
- device type,
- treatment group.

3.5 Ordinal Variables

Ordinal variables have an intrinsic ordering:

$$\text{Low} < \text{Medium} < \text{High}. \quad (10)$$

The numerical distance between categories is not necessarily meaningful.

3.6 Binary Variables

A binary variable has two possible states:

$$X \in \{0, 1\}. \quad (11)$$

Examples include

- success/failure,
- yes/no,
- positive/negative.

4 Step 1: Dataset Audit

The first stage should be a basic structural audit.

Important quantities include

$$N = \text{number of observations}, \quad (12)$$

$$d = \text{number of features}. \quad (13)$$

In Python:

```
1
2 import pandas as pd
3 import numpy as np
4
5 df = pd.read_csv("data.csv")
6
7 print(df.shape)
8 print(df.head())
9 print(df.tail())
10 print(df.info())
11 print(df.describe(include="all"))
```

The following questions should be answered immediately:

- How many rows are present?
- How many columns are present?
- Which columns are numerical?
- Which columns are categorical?
- Which columns contain missing values?
- Which columns contain duplicate information?
- Which column is the target?
- Are there identifier columns?

5 Shape of the Dataset

The dimensionality is

$$X \in \mathbb{R}^{N \times d}. \quad (14)$$

The ratio

$$\frac{d}{N} \quad (15)$$

can be informative.

If

$$d \ll N, \quad (16)$$

the dataset has many observations per feature.

If

$$d \approx N, \quad (17)$$

the problem may be statistically difficult.

If

$$d \gg N, \quad (18)$$

the problem is high-dimensional and overfitting becomes a major concern.

6 Data Types

In Pandas:

```
1  
2 print(df.dtypes)
```

Typical types include

- integer,
- floating point,

- boolean,
- object/string,
- categorical,
- datetime.

The stored data type does not necessarily correspond to the statistical meaning of the variable.

For example,

$$\text{Gender} = \{0, 1\} \quad (19)$$

may be numerically encoded but is categorical rather than continuous.

7 Unique Values

For each variable X_j , define

$$U_j = |\{x_{1j}, x_{2j}, \dots, x_{Nj}\}|. \quad (20)$$

In Python:

```

1 for column in df.columns:
2     print(column)
3     print(df[column].nunique())
4     print(df[column].unique()[:20])
5 
```

A variable with very few unique values may be categorical.

A variable with almost N unique values may instead be continuous or an identifier.

8 Identifiers

Identifier columns require special attention.

For example,

$$ID = \{1, 2, 3, \dots, N\}. \quad (21)$$

An identifier usually contains no predictive information.

Including it in a model can produce spurious relationships.

A useful test is

$$U_j \approx N. \quad (22)$$

If almost every observation has a unique value, the variable should be investigated as a possible identifier.

9 Missing Data

Missing observations are one of the most important components of EDA.

Let

$$M_{ij} = \begin{cases} 1, & x_{ij} \text{ is missing,} \\ 0, & x_{ij} \text{ is observed.} \end{cases} \quad (23)$$

The missingness rate of feature j is

$$r_j = \frac{1}{N} \sum_{i=1}^N M_{ij}. \quad (24)$$

The percentage is

$$100r_j\%. \quad (25)$$

In Python:

```
1 missing_count = df.isnull().sum()
2
3 missing_percentage = (
4     df.isnull().mean() * 100
5 )
6
7
8 print(missing_count)
9 print(missing_percentage)
```

10 Mechanisms of Missingness

Missing data can be classified as

1. MCAR,
2. MAR,
3. MNAR.

10.1 Missing Completely at Random

MCAR means

$$P(M|X, Y) = P(M). \quad (26)$$

Missingness is independent of observed and unobserved values.

10.2 Missing at Random

MAR means missingness depends on observed variables:

$$P(M|X, Y) = P(M|X). \quad (27)$$

10.3 Missing Not at Random

MNAR means missingness depends on the unobserved value itself:

$$P(M|X,Y) \neq P(M|X). \quad (28)$$

This distinction is important because blindly replacing all missing values with the mean can introduce bias.

11 Missing-Value Visualization

Missingness should be visualized.

```
1 import matplotlib.pyplot as plt
2
3 plt.figure(figsize=(10,5))
4
5 plt.imshow(
6     df.isnull(),
7     aspect="auto"
8 )
9
10
11 plt.xlabel("Features")
12 plt.ylabel("Observations")
13 plt.title("Missing Data Pattern")
14
15 plt.show()
```

The goal is not merely to count missing values but to determine whether missingness has structure.

12 Handling Missing Numerical Data

Common approaches include:

- deletion,
- mean imputation,
- median imputation,
- interpolation,
- KNN imputation,
- regression imputation,
- multiple imputation.

Mean imputation is

$$x_{ij} \leftarrow \frac{1}{n_j} \sum_{i: x_{ij} \neq \text{NaN}} x_{ij}. \quad (29)$$

Median imputation replaces missing values by

$$x_{ij} \leftarrow \text{median}(X_j). \quad (30)$$

Median imputation is generally more robust to extreme values.

13 Duplicate Observations

Duplicate rows can distort statistics.

The number of duplicated rows is

```
1  
2 print(df.duplicated().sum())
```

Duplicates can be removed using

```
1  
2 df = df.drop_duplicates()
```

However, duplicates should not automatically be removed.

Repeated measurements may represent legitimate repeated observations.

14 Descriptive Statistics

For a numerical variable

$$X = \{x_1, x_2, \dots, x_N\}, \quad (31)$$

important descriptive statistics include

- mean,
- median,
- variance,
- standard deviation,
- minimum,
- maximum,
- quartiles,
- skewness,
- kurtosis.

15 Mean

The sample mean is

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i. \quad (32)$$

The mean is sensitive to extreme observations.

For example,

$$X = \{1, 2, 3, 4, 100\} \quad (33)$$

has

$$\bar{x} = 22. \quad (34)$$

The mean therefore does not necessarily represent a typical observation.

16 Median

The median is the middle ordered observation.

For odd N ,

$$\text{median}(X) = x_{\frac{N+1}{2}}. \quad (35)$$

For even N ,

$$\text{median}(X) = \frac{x_{\frac{N}{2}} + x_{\frac{N}{2}+1}}{2}. \quad (36)$$

The median is substantially more robust to outliers than the mean.

17 Variance and Standard Deviation

Sample variance is

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2. \quad (37)$$

Standard deviation is

$$s = \sqrt{s^2}. \quad (38)$$

Variance measures the spread around the mean.

18 Quantiles

The p -quantile Q_p satisfies

$$P(X \leq Q_p) \approx p. \quad (39)$$

Important values are

$$Q_{0.25} = Q_1, \quad (40)$$

$$Q_{0.50} = Q_2, \quad (41)$$

$$Q_{0.75} = Q_3. \quad (42)$$

The interquartile range is

$$\boxed{IQR = Q_3 - Q_1}. \quad (43)$$

19 Five-Number Summary

A standard summary consists of

$$\boxed{\{\min, Q_1, Q_2, Q_3, \max\}}. \quad (44)$$

In Python:

```

1
2 print(
3     df.describe()
4 )

```

20 Skewness

Skewness measures asymmetry.

A common sample estimator is

$$\gamma_1 = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^3}{s^3}. \quad (45)$$

Interpretation:

Skewness	Distribution	Typical shape
$\gamma_1 \approx 0$	Symmetric	Balanced
$\gamma_1 > 0$	Right-skewed	Long right tail
$\gamma_1 < 0$	Left-skewed	Long left tail

21 Kurtosis

Kurtosis measures tail heaviness.

A common definition is

$$\gamma_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}. \quad (46)$$

Excess kurtosis is

$$\gamma_2 - 3. \quad (47)$$

Large positive excess kurtosis indicates heavier tails.

22 Distribution Analysis

The distribution of each important numerical variable should be examined.

Common visualizations include:

- histogram,
- density plot,
- box plot,
- violin plot,
- empirical cumulative distribution function.

23 Histogram

A histogram estimates the distribution by dividing the range into bins.

For bin B_k ,

$$n_k = \sum_{i=1}^N \mathbf{1}(x_i \in B_k). \quad (48)$$

The empirical probability is approximately

$$\hat{P}(B_k) = \frac{n_k}{N}. \quad (49)$$

Python:

```
1 import matplotlib.pyplot as plt
2
3
4 df["feature"].hist(
5     bins=30
6 )
7
8 plt.xlabel("Feature")
9 plt.ylabel("Frequency")
10 plt.title("Distribution")
11
12 plt.show()
```

24 Kernel Density Estimation

Kernel Density Estimation (KDE) estimates a continuous probability density.

$$\hat{f}_h(x) = \frac{1}{Nh} \sum_{i=1}^N K\left(\frac{x - x_i}{h}\right) \quad (50)$$

where h is the bandwidth.

For a Gaussian kernel,

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}. \quad (51)$$

KDE is useful for understanding the shape of a distribution.

25 Empirical Cumulative Distribution Function

The empirical CDF is

$$\hat{F}(x) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(x_i \leq x). \quad (52)$$

The ECDF is useful for comparing distributions without choosing histogram bins.

26 Box Plot

The classical box plot is based on

$$Q_1, \quad Q_2, \quad Q_3. \quad (53)$$

The standard Tukey outlier boundaries are

$$L = Q_1 - 1.5IQR, \quad (54)$$

$$U = Q_3 + 1.5IQR. \quad (55)$$

Observations satisfying

$$x < L \quad (56)$$

or

$$x > U \quad (57)$$

are flagged as potential outliers.

Importantly, this is an outlier detection rule, not proof that an observation is erroneous.

27 Categorical EDA

For categorical variables, frequency analysis is essential.

For category c_j ,

$$n_j = \sum_{i=1}^N \mathbf{1}(x_i = c_j). \quad (58)$$

The relative frequency is

$$p_j = \frac{n_j}{N}. \quad (59)$$

The frequencies satisfy

$$\sum_{j=1}^K p_j = 1. \quad (60)$$

Python:

```
1 print(  
2     df["category"].value_counts()  
3 )  
4  
5  
6 print(  
7     df["category"].value_counts(  
8         normalize=True  
9     )  
10 )
```


28 Class Imbalance

For a classification target,

$$Y \in \{0, 1, \dots, K - 1\}, \quad (61)$$

the class proportions are

$$p_k = \frac{N_k}{N}. \quad (62)$$

A highly imbalanced classification problem may have

$$p_0 \gg p_1. \quad (63)$$

This affects both EDA and model evaluation.

Accuracy alone can become misleading.

29 Target Variable Analysis

The target variable must be analyzed separately.

For regression:

$$y \in \mathbb{R}. \quad (64)$$

For classification:

$$y \in \{c_1, \dots, c_K\}. \quad (65)$$

For regression, examine

- distribution,
- range,
- skewness,
- outliers,
- zero values,
- multimodality.

For classification, examine

- class counts,
- class proportions,
- rare classes,
- possible label errors.

30 Bivariate Analysis

Bivariate analysis studies relationships between two variables.

Examples include

$$X \leftrightarrow Y, \quad (66)$$

or

$$X_i \leftrightarrow X_j. \quad (67)$$

Appropriate tools depend on variable types.

Variable pair	Useful tools
Numerical–Numerical	Scatter plot, correlation, regression
Categorical–Numerical	Box plot, violin plot, ANOVA
Categorical–Categorical	Contingency table, χ^2 test
Numerical–Target	Scatter plot, correlation
Categorical–Target	Group statistics, box plot

31 Covariance

For variables X and Y ,

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (68)$$

The sample covariance is

$$s_{XY} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}). \quad (69)$$

Positive covariance means the variables tend to increase together.

Negative covariance means one tends to increase as the other decreases.

32 Pearson Correlation

Pearson's correlation coefficient is

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}. \quad (70)$$

The sample version is

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}. \quad (71)$$

It satisfies

$$-1 \leq r \leq 1. \quad (72)$$

Interpretation:

- $r \approx 1$: strong positive linear relationship,
- $r \approx -1$: strong negative linear relationship,

- $r \approx 0$: little linear association.

A crucial limitation is

$$r \approx 0 \not\Rightarrow X \perp Y. \quad (73)$$

A nonlinear relationship can have near-zero Pearson correlation.

33 Spearman Correlation

Spearman correlation measures monotonic association using ranks.

If R_i and S_i are ranks,

$$\rho_s = 1 - \frac{6 \sum_i d_i^2}{N(N^2 - 1)}, \quad (74)$$

when there are no tied ranks.

It is often more appropriate than Pearson correlation for nonlinear monotonic relationships and variables containing strong outliers.

34 Scatter Plots

A scatter plot displays

$$(x_i, y_i). \quad (75)$$

Python:

```

1
2 import matplotlib.pyplot as plt
3
4 plt.scatter(
5     df["feature"],
6     df["target"]
7 )
8
9 plt.xlabel("Feature")
10 plt.ylabel("Target")
11 plt.title("Feature vs Target")
12
13 plt.show()
```

Scatter plots can reveal

- linearity,
- nonlinear relationships,
- clusters,
- heteroscedasticity,
- outliers,
- saturation,
- discontinuities.

35 Nonlinear Relationships

Consider

$$Y = X^2. \quad (76)$$

If X is symmetric around zero, Pearson correlation can be close to zero even though

$$Y \quad (77)$$

is completely determined by X .

Therefore,

Correlation $\neq$ general dependence.

(78)

Useful nonlinear dependence measures include

- mutual information,
- distance correlation,
- kernel dependence measures.

36 Mutual Information

Mutual information is

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$$

(79)

for discrete variables.

For continuous variables,

$$I(X; Y) = \int \int p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx dy. \quad (80)$$

It measures general statistical dependence rather than only linear dependence.

37 Correlation Matrix

For features

$$X_1, \dots, X_d, \quad (81)$$

the correlation matrix is

$$R = \begin{bmatrix} 1 & r_{12} & \cdots & r_{1d} \\ r_{21} & 1 & \cdots & r_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ r_{d1} & r_{d2} & \cdots & 1 \end{bmatrix}. \quad (82)$$

Python:

```

1 corr = df.select_dtypes(
2     include=np.number
3 ).corr()
4
5
6 print(corr)

```

A heatmap can make the structure easier to inspect.

38 Multicollinearity

Multicollinearity occurs when predictors are strongly related.

For example,

$$X_3 \approx 2X_1 + X_2. \quad (83)$$

This can make linear regression coefficients unstable.

One diagnostic is the Variance Inflation Factor (VIF):

$$VIF_j = \frac{1}{1 - R_j^2}, \quad (84)$$

where R_j^2 is obtained by regressing feature X_j on all the other predictors.
As

$$R_j^2 \rightarrow 1, \quad (85)$$

we obtain

$$VIF_j \rightarrow \infty. \quad (86)$$

39 Categorical–Numerical Relationships

Suppose X is categorical and Y is numerical.

For category k ,

$$\bar{y}_k = \frac{1}{N_k} \sum_{i:x_i=k} y_i. \quad (87)$$

Compare

$$\bar{y}_1, \bar{y}_2, \dots, \bar{y}_K. \quad (88)$$

Useful plots include

- box plots,
- violin plots,
- strip plots,
- grouped bar plots.

40 Categorical–Categorical Relationships

For categorical variables X and Y , construct a contingency table:

$$O_{ij} = \#\{X = c_i, Y = d_j\}. \quad (89)$$

The expected count under independence is

$$E_{ij} = \frac{(\text{row total}_i)(\text{column total}_j)}{N}. \quad (90)$$

The Pearson χ^2 statistic is

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}. \quad (91)$$

41 Outlier Analysis

An outlier is an observation that is unusually distant from the majority of observations.

Outliers can be

- genuine rare events,
- measurement errors,
- data-entry errors,
- sensor failures,
- distributional extremes.

EDA should identify outliers rather than automatically delete them.

42 Z-Score Method

The standardized value is

$$z_i = \frac{x_i - \bar{x}}{s}. \quad (92)$$

A common heuristic flags

$$|z_i| > 3. \quad (93)$$

However, this assumes that the mean and standard deviation provide a meaningful description of the distribution.

For skewed distributions, this approach can be inappropriate.

43 Robust Outlier Detection

The median and MAD provide a robust alternative.

The Median Absolute Deviation is

$$MAD = \text{median}(|x_i - \text{median}(X)|). \quad (94)$$

A robust standardized score can be defined as

$$z_i^* = \frac{x_i - \text{median}(X)}{1.4826 MAD}. \quad (95)$$

44 Multivariate Outliers

An observation can be normal in every individual feature but unusual in the joint feature space.
The Mahalanobis distance is

$$D_M(x) = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}. \quad (96)$$

This accounts for covariance between features.
For example, the point

$$(x_1, x_2) \quad (97)$$

may not be extreme individually but may violate the relationship

$$x_2 \approx f(x_1). \quad (98)$$

45 Time-Series EDA

Time-series data requires additional analysis because observations are not independent.
Given

$$\{x_t\}_{t=1}^T, \quad (99)$$

inspect:

- trend,
- seasonality,
- autocorrelation,
- stationarity,
- structural breaks,
- volatility,
- missing intervals.

46 Trend

A time series can be decomposed conceptually as

$$x_t = T_t + S_t + R_t, \quad (100)$$

where

- T_t is the trend,
- S_t is seasonality,
- R_t is the residual component.

For multiplicative decomposition,

$$x_t = T_t S_t R_t. \quad (101)$$

47 Autocorrelation

The autocorrelation function is

$$\rho_k = \frac{\text{Cov}(X_t, X_{t-k})}{\text{Var}(X_t)}. \quad (102)$$

The sample autocorrelation at lag k is

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^T (x_t - \bar{x})^2}. \quad (103)$$

High autocorrelation indicates temporal dependence.

48 Stationarity

A weakly stationary process satisfies

$$\mathbb{E}[X_t] = \mu, \quad (104)$$

$$\text{Var}(X_t) = \sigma^2, \quad (105)$$

and

$$\text{Cov}(X_t, X_{t-k}) = \gamma_k, \quad (106)$$

independent of absolute time t .

Nonstationarity may arise from

- trend,
- changing variance,
- structural changes,
- seasonality.

49 Differencing

First differencing is

$$\Delta X_t = X_t - X_{t-1}. \quad (107)$$

Second differencing is

$$\Delta^2 X_t = \Delta X_t - \Delta X_{t-1}. \quad (108)$$

Differencing can remove certain forms of trend.

50 Feature Scaling

Scaling is important for many machine-learning algorithms.

Suppose

$$X_j \in [a_j, b_j]. \quad (109)$$

One common transformation is standardization:

$$z = \frac{x - \mu}{\sigma}. \quad (110)$$

After standardization,

$$\mathbb{E}[Z] \approx 0, \quad (111)$$

and

$$\text{Var}(Z) \approx 1. \quad (112)$$

51 Min-Max Scaling

Min-max scaling maps the feature to $[0, 1]$:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \quad (113)$$

More generally, to $[a, b]$:

$$x' = a + (b - a) \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \quad (114)$$

52 Robust Scaling

Robust scaling uses the median and IQR:

$$x' = \frac{x - \text{median}(X)}{IQR}. \quad (115)$$

This is less sensitive to extreme observations.

53 When Scaling Matters

Scaling is especially important for algorithms based on distances or gradients.

Examples:

- KNN,
- K-means,
- SVM,
- PCA,
- Logistic Regression,

- neural networks.

Tree-based models such as Random Forests generally do not require feature scaling because splits depend on feature ordering rather than Euclidean distances.

54 Transformation of Variables

EDA may reveal that a variable is strongly skewed.

A transformation can make the distribution more suitable for modelling.

55 Log Transformation

For positive data,

$$x' > 0, \quad (116)$$

use

$$\boxed{x' = \log(x)}. \quad (117)$$

For data containing zero,

$$x' = \log(1 + x) \quad (118)$$

is often used.

Log transformations are useful when

- values span several orders of magnitude,
- the distribution is strongly right-skewed,
- multiplicative relationships are suspected.

56 Power Transformations

The Box–Cox transformation is

$$x^{(\lambda)} = \begin{cases} \frac{x^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log x, & \lambda = 0. \end{cases} \quad (119)$$

It is applicable to positive observations.

The Yeo–Johnson transformation extends this idea to zero and negative values.

57 Feature Engineering

EDA frequently reveals useful transformations.

Given features

$$X_1, X_2, \dots, X_d, \quad (120)$$

we may construct

$$Z = X_1 X_2, \quad (121)$$

$$Z = \frac{X_1}{X_2}, \quad (122)$$

or

$$Z = X_1^2. \quad (123)$$

However, feature engineering must be based on domain knowledge and validated statistically.

58 Polynomial Features

A polynomial model can include

$$x, \quad x^2, \quad x^3, \dots \quad (124)$$

For two features,

$$X_1, X_2, \quad (125)$$

second-order features include

$$X_1^2, \quad X_2^2, \quad X_1 X_2. \quad (126)$$

Polynomial features can allow linear models to represent nonlinear relationships.

59 Interaction Effects

An interaction occurs when the effect of one variable depends on another.

A regression model can contain

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \epsilon. \quad (127)$$

The term

$$X_1 X_2 \quad (128)$$

represents the interaction.

60 Dimensionality Reduction

When

$$d \gg 1, \quad (129)$$

EDA may require dimensionality reduction.

Important methods include

- PCA,
- Kernel PCA,
- t-SNE,
- UMAP,
- autoencoders.

61 Principal Component Analysis

First center the data:

$$X_c = X - \mathbf{1}\mu^T. \quad (130)$$

The covariance matrix is

$$\Sigma = \frac{1}{N-1} X_c^T X_c. \quad (131)$$

PCA solves

$$\Sigma v_k = \lambda_k v_k. \quad (132)$$

The principal components are ordered such that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d. \quad (133)$$

The explained variance ratio is

$$EV R_k = \frac{\lambda_k}{\sum_j \lambda_j}. \quad (134)$$

The cumulative explained variance is

$$CEV_m = \sum_{k=1}^m EV R_k. \quad (135)$$

62 t-SNE

t-SNE attempts to preserve local neighborhood structure when mapping high-dimensional data into a low-dimensional space.

It defines similarities in the original space and low-dimensional space and minimizes a KL divergence:

$$C = KL(P\|Q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}}. \quad (136)$$

t-SNE is primarily a visualization method rather than a general-purpose feature extraction algorithm.

63 UMAP

UMAP constructs a neighborhood graph in the high-dimensional space and attempts to preserve its structure in a lower-dimensional representation.

UMAP is often useful for visualizing clusters and nonlinear manifolds.

However, apparent clusters should not automatically be interpreted as physically or statistically meaningful classes.

64 Class Separation

For classification, EDA should examine whether classes occupy different regions of feature space.

For classes $C_1, \dots, C_K$, inspect

$$p(X|C_k). \quad (137)$$

If

$$p(X|C_1) \approx p(X|C_2), \quad (138)$$

classification may be intrinsically difficult.

Useful visualizations include

- pair plots,
- PCA projections,
- UMAP,
- class-conditional histograms,
- feature distributions by class.

65 Target Leakage

Data leakage occurs when information unavailable at prediction time is used during training.

Suppose the target is

$$Y. \quad (139)$$

If a feature contains information generated after Y , then

$$X_{\text{future}} \rightarrow Y \quad (140)$$

may produce unrealistically high model performance.

EDA must therefore examine the meaning and temporal origin of every feature.

66 Train-Test Leakage

Preprocessing should generally be fitted only on the training set.

Incorrect:

$$\text{Scale entire dataset} \rightarrow \text{Train-test split}. \quad (141)$$

Correct:

$$\text{Train-test split} \rightarrow \text{Fit scaler on train} \rightarrow \text{Transform train and test}. \quad (142)$$

For example:

```
1
2 from sklearn.model_selection import train_test_split
3 from sklearn.preprocessing import StandardScaler
4
5 X_train, X_test, y_train, y_test = (
6     train_test_split(
```

```

7         X,
8         y,
9         test_size=0.2,
10        random_state=42
11    )
12 )
13
14 scaler = StandardScaler()
15
16 X_train = scaler.fit_transform(X_train)
17 X_test = scaler.transform(X_test)

```

67 Train–Validation–Test Structure

A supervised dataset is commonly divided into

$$\mathcal{D} = \mathcal{D}_{train} \cup \mathcal{D}_{val} \cup \mathcal{D}_{test}. \quad (143)$$

The training set estimates model parameters,

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{train}(\theta). \quad (144)$$

The validation set is used for hyperparameter selection.

The test set is reserved for final evaluation.

The test set should not influence model development.

68 EDA and Cross-Validation

For small datasets, cross-validation can provide more stable estimates.

In K -fold cross-validation,

$$\mathcal{D} = D_1 \cup D_2 \cup \dots \cup D_K. \quad (145)$$

At iteration k ,

$$D_k \quad (146)$$

is used for validation while the remaining folds are used for training.

The estimated score is

$$\bar{S} = \frac{1}{K} \sum_{k=1}^K S_k. \quad (147)$$

EDA should respect the data-generating structure when creating folds.

For time series, random shuffling can create leakage and should generally be avoided.

69 Distribution Shift

A training distribution may differ from the deployment distribution.

Let

$$P_{train}(X) \neq P_{test}(X). \quad (148)$$

This is covariate shift.

More generally,

$$P_{train}(X, Y) \neq P_{test}(X, Y). \quad (149)$$

EDA should compare training and deployment-like samples whenever possible.

70 Population and Sampling

A dataset is usually a sample from a larger population.

Let the population distribution be

$$P(X, Y). \quad (150)$$

The observed dataset consists of

$$(X_i, Y_i) \sim P(X, Y). \quad (151)$$

EDA should investigate whether the observed sample reasonably represents the intended population.

Sampling bias occurs when

$$P_{sample}(X) \neq P_{population}(X). \quad (152)$$

71 Visualization Strategy

A systematic EDA visualization strategy is more useful than producing plots randomly.

71.1 Univariate Numerical

Use

- histogram,
- KDE,
- box plot,
- ECDF.

71.2 Univariate Categorical

Use

- count plot,
- frequency table.

71.3 Numerical–Numerical

Use

- scatter plot,
- hexbin plot,
- correlation analysis.

71.4 Categorical–Numerical

Use

- box plot,
- violin plot,
- grouped statistics.

71.5 Categorical–Categorical

Use

- contingency table,
- stacked bar plot,
- heatmap.

72 Pair Plot

A pair plot displays pairwise relationships among numerical variables and their marginal distributions.

It can reveal

- linear relationships,
- nonlinear relationships,
- clusters,
- outliers,
- heteroscedasticity.

For large d , however, pair plots become difficult to interpret.

73 Heteroscedasticity

Heteroscedasticity occurs when

$$\text{Var}(Y|X = x) \tag{153}$$

depends on x .

For example,

$$\text{Var}(Y|X = x) \propto x^2. \tag{154}$$

A scatter plot may show a funnel-shaped pattern.

This matters particularly for statistical regression models.

74 Residual EDA

After fitting a model,

$$\hat{y}_i = f_{\theta}(x_i), \quad (155)$$

define residuals

$$e_i = y_i - \hat{y}_i. \quad (156)$$

Residual analysis should investigate

- residual mean,
- residual variance,
- residual distribution,
- residual autocorrelation,
- residual dependence on predictors.

An idealized residual structure is

$$\mathbb{E}[e|X] = 0. \quad (157)$$

75 Normality

A normal distribution has density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right]. \quad (158)$$

Normality can be assessed using

- histogram,
- Q–Q plot,
- Shapiro–Wilk test,
- Anderson–Darling test.

However, normality of raw features is not universally required for machine learning.

It matters primarily when the statistical model or inference procedure makes such assumptions.

76 Q–Q Plot

A Q–Q plot compares empirical quantiles with theoretical quantiles.

If

$$Q_{emp}(p) \approx Q_{theory}(p), \quad (159)$$

the points approximately follow a straight line.

Deviations in the tails can reveal

- heavy tails,
- skewness,
- outliers.

77 EDA for Regression

For regression problems,

$$Y \in \mathbb{R}, \tag{160}$$

EDA should examine:

1. target distribution,
2. feature-target relationships,
3. feature-feature relationships,
4. outliers,
5. heteroscedasticity,
6. nonlinearities,
7. transformations,
8. missingness,
9. potential leakage.

Important plots include

- X_j versus Y scatter plots,
- target histogram,
- residual plots,
- correlation matrix,
- feature distributions.

78 EDA for Classification

For classification,

$$Y \in \{1, \dots, K\}. \tag{161}$$

EDA should investigate

- class balance,
- class-conditional distributions,
- separability,
- outliers,

- missingness by class,
- categorical relationships,
- leakage.

For each feature,

$$p(X_j|Y = c_k) \tag{162}$$

can be compared across classes.

79 EDA for Time Series

For time series, the EDA workflow should include

Time ordering → Trend → Seasonality → Autocorrelation → Stationarity → Anomalies → Forecasting structure
--

(163)

Random train-test splitting should generally be avoided when it allows future information to enter the training data.

80 EDA for Scientific Data

Scientific data requires additional domain-aware analysis.

Suppose

$$y = f(x; \theta). \tag{164}$$

EDA should determine whether the measured quantities obey expected physical relationships. Examples include

$$E = mc^2, \tag{165}$$

$$PV = nRT, \tag{166}$$

or conservation laws such as

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \tag{167}$$

EDA can therefore involve checking

- dimensional consistency,
- conservation laws,
- physical bounds,
- symmetry,
- monotonicity,
- known scaling relationships,
- measurement uncertainty.

81 Dimensional Analysis

Physical quantities have dimensions.

For example,

$$[v] = LT^{-1}, \quad (168)$$

$$[a] = LT^{-2}, \quad (169)$$

and

$$[F] = MLT^{-2}. \quad (170)$$

A physically meaningful equation must be dimensionally consistent.

For example,

$$F = ma \quad (171)$$

has

$$[F] = [M][L][T^{-2}]. \quad (172)$$

EDA of scientific datasets should therefore check units before statistical modelling.

82 Physical Bounds

Some quantities have natural bounds.

For a probability,

$$0 \leq p \leq 1. \quad (173)$$

For a concentration,

$$c \geq 0. \quad (174)$$

For absolute temperature,

$$T \geq 0 \text{ K}. \quad (175)$$

Values violating known physical constraints may indicate measurement or preprocessing errors.

83 Uncertainty Analysis

Scientific measurements frequently have uncertainty:

$$x = x_{\text{true}} + \epsilon. \quad (176)$$

A measurement may be represented as

$$x \pm \sigma_x. \quad (177)$$

If

$$y = f(x), \quad (178)$$

first-order uncertainty propagation gives

$$\sigma_y^2 \approx \left(\frac{df}{dx} \right)^2 \sigma_x^2. \quad (179)$$

For multiple variables,

$$\sigma_y^2 \approx \nabla f^T \Sigma \nabla f. \quad (180)$$

84 EDA for Neural Operators

Neural-operator datasets have a special structure.

Instead of

$$x_i \in \mathbb{R}^d, \quad (181)$$

we may have

$$a_i : \Omega \rightarrow \mathbb{R}^m, \quad (182)$$

and

$$u_i : \Omega \rightarrow \mathbb{R}^q. \quad (183)$$

Thus

$$\mathcal{D} = \{(a_i, u_i)\}_{i=1}^N. \quad (184)$$

EDA must examine not only individual scalar features but entire fields.

Important quantities include

- spatial resolution,
- domain geometry,
- boundary conditions,
- initial conditions,
- coefficient distributions,
- field magnitude,
- spatial smoothness,
- spectral energy,
- solution norms,
- parameter distributions.

85 Field Statistics

For a field

$$u(x), \quad (185)$$

the spatial mean is

$$\bar{u} = \frac{1}{|\Omega|} \int_{\Omega} u(x) dx. \quad (186)$$

The spatial variance is

$$\sigma_u^2 = \frac{1}{|\Omega|} \int_{\Omega} (u(x) - \bar{u})^2 dx. \quad (187)$$

The L^2 norm is

$$\|u\|_{L^2(\Omega)} = \left(\int_{\Omega} |u(x)|^2 dx \right)^{1/2}. \quad (188)$$

For discretized fields,

$$\|u\|_2 = \left(\sum_i |u_i|^2 \right)^{1/2}. \quad (189)$$

86 Spectral EDA

For spatial fields, Fourier analysis can reveal characteristic scales.

The Fourier transform is

$$\hat{u}(k) = \int u(x) e^{-ikx} dx. \quad (190)$$

The spectral energy can be represented by

$$E(k) = |\hat{u}(k)|^2. \quad (191)$$

This is particularly important for neural operators such as FNOs.

If most energy occurs at low frequencies,

$$E(k) \approx 0 \quad \text{for large } |k|, \quad (192)$$

a low-frequency representation may capture most of the field structure.

87 Spatial Correlation

A spatial covariance function can be defined as

$$C(r) = \mathbb{E}[u(x)u(x+r)]. \quad (193)$$

The corresponding correlation length provides information about the characteristic spatial scale.

This can influence

- grid resolution,

- convolution kernel size,
- Fourier mode selection,
- neural-operator architecture.

88 PDE Residual EDA

If the governing equation is known,

$$\mathcal{F}[u] = 0, \tag{194}$$

EDA can evaluate the residual of the dataset itself:

$$r(x) = \mathcal{F}[u](x). \tag{195}$$

For high-quality numerical solutions,

$$\|r\| \tag{196}$$

should generally be small relative to the scale of the problem.
This provides a physics-based data-quality check.

89 Complete EDA Workflow

A comprehensive EDA workflow can be organized as follows.

1. Dataset Audit

Shape, types, columns, target, identifiers

↓

2. Data Quality

Missing values, duplicates, invalid values

↓

3. Univariate Analysis

Mean, median, variance, distributions, outliers

↓

4. Bivariate Analysis

Correlation, scatter plots, group comparisons

↓

5. Multivariate Analysis

Correlation structure, PCA, clustering

↓

6. Target Analysis

Class balance or regression structure

↓

7. Leakage Analysis

Temporal leakage, target leakage, preprocessing leakage

↓

8. Feature Engineering

Transformation, scaling, interactions

↓

9. Model-Relevant Analysis

Linearity, dimensionality, temporal/spatial structure

↓

10. Final Dataset

Clean, validated, model-ready data

(197)

90 A Practical Python EDA Template

The following provides a compact but comprehensive starting point.

```
1
2 import numpy as np
3 import pandas as pd
4 import matplotlib.pyplot as plt
5
6 # -----
```



```

7 # 1. Load data
8 # -----
9
10 df = pd.read_csv("data.csv")
11
12 # -----
13 # 2. Basic structure
14 # -----
15
16 print("Shape:")
17 print(df.shape)
18
19 print("\nColumns:")
20 print(df.columns)
21
22 print("\nData types:")
23 print(df.dtypes)
24
25 print("\nFirst rows:")
26 print(df.head())
27
28 print("\nSummary:")
29 print(df.describe(include="all"))
30
31 # -----
32 # 3. Missing values
33 # -----
34
35 missing = pd.DataFrame({
36     "count": df.isnull().sum(),
37     "percentage": df.isnull().mean() * 100
38 })
39
40 print("\nMissing values:")
41 print(missing)
42
43 # -----
44 # 4. Duplicates
45 # -----
46
47 print(
48     "\nDuplicate rows:",
49     df.duplicated().sum()
50 )
51
52 # -----
53 # 5. Unique values
54 # -----
55
56 for column in df.columns:
57
58     print(
59         f"\n{column}: ",
60         f"{df[column].nunique()} unique values"
61     )
62
63 # -----
64 # 6. Numerical and categorical columns

```

```

65 # -----
66
67 numerical = df.select_dtypes(
68     include=np.number
69 ).columns
70
71 categorical = df.select_dtypes(
72     exclude=np.number
73 ).columns
74
75 print("\nNumerical:")
76 print(list(numerical))
77
78 print("\nCategorical:")
79 print(list(categorical))
80
81 # -----
82 # 7. Numerical distributions
83 # -----
84
85 for column in numerical:
86
87     plt.figure()
88
89     df[column].hist(
90         bins=30
91     )
92
93     plt.xlabel(column)
94     plt.ylabel("Frequency")
95     plt.title(
96         f"Distribution of {column}"
97     )
98
99     plt.show()
100
101 # -----
102 # 8. Box plots
103 # -----
104
105 for column in numerical:
106
107     plt.figure()
108
109     plt.boxplot(
110         df[column].dropna()
111     )
112
113     plt.ylabel(column)
114     plt.title(
115         f"Box Plot of {column}"
116     )
117
118     plt.show()
119
120 # -----
121 # 9. Categorical frequencies
122 # -----

```

```

123
124 for column in categorical:
125
126     print(
127         f"\n{column}"
128     )
129
130     print(
131         df[column].value_counts(
132             dropna=False
133         )
134     )
135
136 # -----
137 # 10. Correlation
138 # -----
139
140 corr = df[
141     numerical
142 ].corr()
143
144 print("\nCorrelation matrix:")
145 print(corr)
146
147 # -----
148 # 11. Target relationship
149 # -----
150
151 target = "target"
152
153 for column in numerical:
154
155     if column != target:
156
157         plt.figure()
158
159         plt.scatter(
160             df[column],
161             df[target]
162         )
163
164         plt.xlabel(column)
165         plt.ylabel(target)
166         plt.title(
167             f"{column} vs {target}"
168         )
169
170         plt.show()

```

91 EDA Checklist

Before beginning modelling, the following checklist should be completed.

Dataset Structure

- ☐ Number of observations determined.

- ☐ Number of features determined.
- ☐ Target identified.
- ☐ Numerical variables identified.
- ☐ Categorical variables identified.
- ☐ Identifier columns identified.

Data Quality

- ☐ Missing values quantified.
- ☐ Missingness patterns investigated.
- ☐ Duplicate observations checked.
- ☐ Invalid values checked.
- ☐ Physical bounds checked where applicable.
- ☐ Units verified.

Statistical Structure

- ☐ Mean and median examined.
- ☐ Variance examined.
- ☐ Quantiles examined.
- ☐ Skewness examined.
- ☐ Outliers investigated.
- ☐ Feature distributions visualized.

Relationships

- ☐ Correlation matrix examined.
- ☐ Nonlinear relationships investigated.
- ☐ Feature-target relationships examined.
- ☐ Multicollinearity investigated.
- ☐ Class-conditional distributions examined.

Machine-Learning Considerations

- ☐ Class imbalance checked.
- ☐ Scaling requirements determined.
- ☐ Transformations considered.
- ☐ Data leakage investigated.
- ☐ Train-test strategy determined.
- ☐ Cross-validation strategy determined.

Scientific Data

- ☐ Units verified.
- ☐ Dimensional consistency checked.
- ☐ Physical bounds checked.
- ☐ Conservation laws considered.
- ☐ Measurement uncertainty examined.
- ☐ Spatial/temporal structure examined.
- ☐ PDE residuals checked when applicable.

92 Common EDA Mistakes

92.1 Mistake 1: Removing Every Outlier

An outlier is not necessarily an error.

A rare observation may be the most scientifically important point in the dataset.

92.2 Mistake 2: Assuming Correlation Means Causation

The relationship

$$X \leftrightarrow Y \tag{198}$$

does not establish

$$X \rightarrow Y. \tag{199}$$

Confounding variables may explain the observed association.

92.3 Mistake 3: Using Only a Correlation Matrix

Pearson correlation detects linear dependence.

It cannot reliably detect all nonlinear relationships.

92.4 Mistake 4: Ignoring the Target

EDA should not analyze only the predictors.

The target distribution and its relationship with the predictors are central to the modelling problem.

92.5 Mistake 5: Scaling Before Splitting

This can leak information from the test set into the training process.

92.6 Mistake 6: Random Splitting Time Series

For temporal data, random splitting may allow future observations to influence training.

92.7 Mistake 7: Blindly Removing Missing Rows

Deletion can introduce selection bias if missingness is systematic.

92.8 Mistake 8: Treating Plots as Proof

Visualization generates hypotheses.

Statistical tests, domain knowledge, and out-of-sample validation are needed before making strong conclusions.

93 EDA and Model Selection

EDA should ultimately inform the choice of model.

For approximately linear relationships,

$$Y \approx \beta_0 + \beta^T X, \quad (200)$$

linear models may be appropriate.

For nonlinear tabular relationships,

$$Y \approx f(X), \quad (201)$$

tree ensembles, kernels, or neural networks may be appropriate.

For temporal dependence,

$$X_t \leftrightarrow X_{t-k}, \quad (202)$$

time-series models should be considered.

For spatial structure,

$$u(x, y), \quad (203)$$

CNNs or spatial neural operators may be appropriate.

For mappings between functions,

$$a(\cdot) \rightarrow u(\cdot), \quad (204)$$

operator-learning architectures become appropriate.

Thus,

EDA $\rightarrow$ Understanding of Data Geometry $\rightarrow$ Inductive Bias $\rightarrow$ Model Choice.

(205)

94 Final Summary

Exploratory Data Analysis is fundamentally an investigation of the statistical structure of the observed data.

At the simplest level,

EDA = Structure + Distribution + Relationships + Quality + Context.

(206)

A complete EDA process should answer five fundamental questions:

1. **What is in the dataset?**
2. **Is the data reliable?**
3. **How are the variables distributed?**
4. **How are the variables related?**

5. What does this imply for modelling?

For ordinary machine learning, EDA determines whether the data is approximately linear, nonlinear, sparse, imbalanced, high-dimensional, temporal, categorical, or otherwise structured.

For scientific machine learning, EDA goes further by examining

$$\boxed{\text{statistical structure} + \text{geometrical structure} + \text{temporal structure} + \text{physical structure.}} \quad (207)$$

For neural-operator datasets in particular, EDA must move beyond ordinary column-wise statistics and examine the structure of the underlying functions and fields:

$$a(x) \rightarrow u(x), \quad (208)$$

including spatial correlations, spectral content, boundary conditions, parameter distributions, resolution, and physical consistency.

Consequently, EDA is not simply a preliminary visualization step. It is the first stage of understanding the mathematical structure of a machine-learning problem.